

AnyJev Technical Report

Typed Decisions from One Prefill of Any Open LLM

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Abstract

A typed decision is a choice among a fixed set of options, returned as a probability rather than as text. Systems that need typed decisions today use models trained for that purpose. This report describes AnyJev, which reads a typed decision from one prefill of a pretrained instruction-tuned language model. The readout restricts the next-token distribution at the answer position to the option tokens. It has two defects: the model assigns higher probability to some labels whatever the input, and to some positions in the option list. AnyJev corrects both with no gradient steps and no parameter changes: it divides out a label prior estimated from unlabelled inputs, and it averages log-probabilities over the K cyclic rotations of the option list. On two 20-option tasks the rotations lower the order-flip rate from 0.33 to 0.14 and from 0.33 to 0.18, and raise accuracy on 11 of 11 models on both. Reading every rotation requires K prefills. A stopping rule selected against the full-rotation decision on unlabelled states cuts that. Selecting the threshold on one unlabelled split and bounding its disagreement on a second, it reads 10.6 rotations of 18 at a verified 0.008 bound on two of four cells; selected and bounded on one split, as our serving run did, it reads 7.3 and serves 2.2 times as many decisions per second on vLLM. The code is open source.

 [nokia-applied-research/AnyJev](https://github.com/nokia-applied-research/AnyJev)

Early report on work in development

1 Introduction

An agent system has to make many small decisions that are not text: route this request to one of twenty skills, decide whether this input is an injection attempt, score this answer from one to five. A language model can be asked for these in prose, but the caller then has to parse the reply, and gets no probability it can threshold. A *typed decision* fixes the interface instead. The caller supplies a state and a question with a fixed set of options, and the system returns a distribution over those options. The interface was made popular by a hosted service, and open implementations now include both trained decision models and frozen-model readouts (Section 2).

This report takes a different position (Figure 1). The capability is already present in the pretrained instruction-tuned models we tested, and what needs correcting is the readout rather than the model. AnyJev builds one prompt from the state and the question, runs a single prefill, and reads the decision from the next-token distribution at the answer position, restricted to the tokens that name the options. No parameter

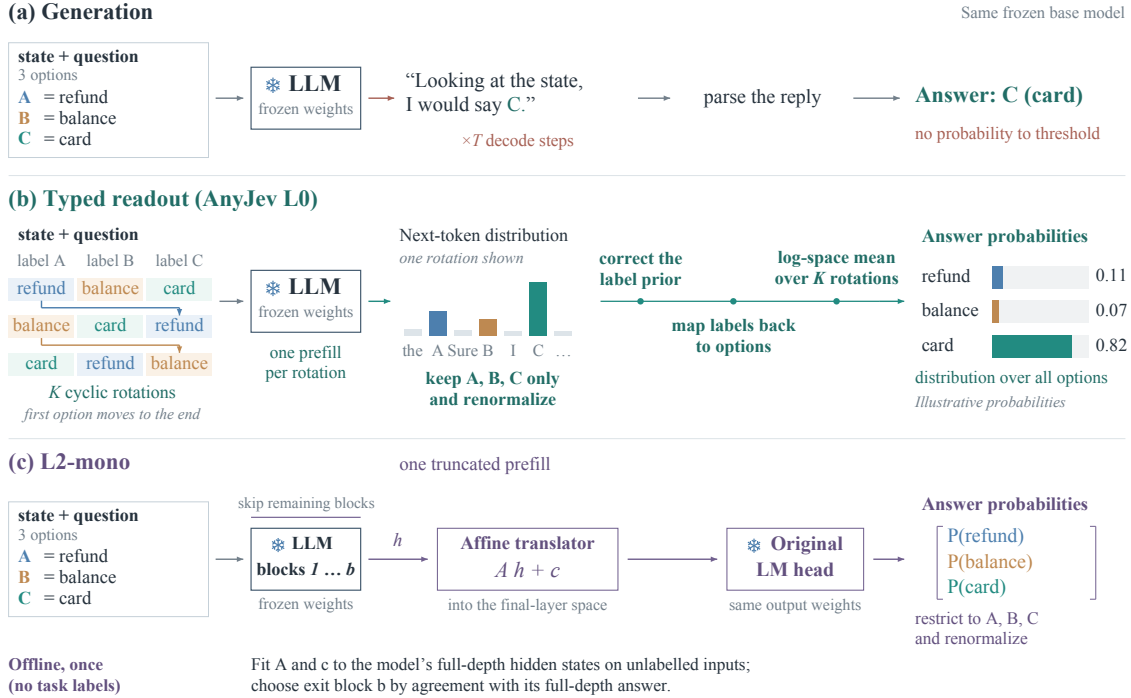


Figure 1. The same base model, three decision paths. (a) Asking for the answer in text costs one forward pass per generated token and returns prose the caller has to parse, with no probability attached. (b) AnyJev L0 runs one prefill per cyclic rotation. It restricts the next-token distribution to the option-label tokens and renormalises, corrects the label prior, maps each position back to its option, and averages in log space across the K rotations before renormalising. A, B and C are position labels whose option assignments change across rotations; the final distribution is indexed by option. Probabilities are illustrative, and one rotation’s token distribution is shown. L0 debiases the readout; confidence calibration is a separate L1 step. (c) L2-mono truncates the forward at a selected block, translates the hidden state into the final-layer space with an affine map, and uses the original output head for the option readout. The translator is fitted once on unlabelled inputs to the model’s full-depth hidden states; the exit depth is selected by agreement with its full-depth answers, without task labels. The L2-mono output is symbolic. All three paths keep the base-model weights fixed.

changes, no generated tokens, and no second model.

Read that way, the distribution is usable and wrong in two specific ways. The model assigns higher probability to some labels whatever the state says, and it assigns higher probability to some positions in the option list. The second defect is the one a caller notices. On a 20-option task, reversing the option list changes the answer on 33% of items, averaged over eleven models. Both defects are properties of the readout rather than of the model’s knowledge, Section 4 corrects both with no gradient steps, from the model’s own outputs on unlabelled inputs.

Accuracy varies with the base model and changes little under the corrections. On the public subset of a typed-decision benchmark the eight models we ran span 0.549 to 0.798, and the debiasing does not change accuracy there by a measurable amount. It supplies a large reduction in order dependence and, with a small labelled set, calibrated confidence. Section 7 separates the two and reports each with the interval it was measured to.

2 Related Work

2.1 Typed decision models and open implementations

Jev popularizes probabilistic Choice, Score, and Noul interfaces [2], building on the broader connection between language-model predictions and classification [40, 15]. Frozen-model implementations include Yinsongxu’s LLM2Jev, which scores candidates independently, tic-top’s llm2jev, which reads labels from a joint option prompt, and SemIf, which supports shared-state execution [31, 32, 41]. NanoJev, Kev, Laya, and related releases explore learned scorers and local decision services [35, 24, 6, 27, 33, 22, 13]. Visual Jev extends shared-context decisions to images, while NumericJev obtains numerical outputs through multi-round interval refinement [52, 51].

2.2 Probability readouts and calibration

Label-probability readouts inherit surface-form and option-order biases [20, 56]. AnyJev combines prior correction related to content-free and batch calibration with temperature scaling [55, 58, 19], while measuring the prefill cost of averaging layouts. Recent Jev studies expose sensitivity to option-name semantics and inconsistency across logically linked questions [43, 29]. These distinguish schema validity from semantic correctness and probability quality; AnyJev focuses on label and position bias and calibrated confidence on frozen backbones.

2.3 Decision components in agent systems

Model cascades and Jev-based judging use confidence to accept or escalate decisions, although correlated errors can limit fallback gains [8, 48, 30, 39]. Open applications use typed decisions for browser control, history pruning, and code retrieval [21, 12, 11]. Such components complement Nokia’s KV-cache compression, context management, and MoE paging work [54, 34, 53]. AnyJev could also support production agents in autonomous networks, motivated by AI-native 6G visions and Jev-based edge orchestration [47, 28]. Self-Evolving Code-with-Image Reasoning and Field-Aware Agent Skill Retrieval suggest roles in solver invocation and skill selection [50, 16]; recommendation and time-series systems provide further domain contexts [45, 57, 9]. These proposed combinations require application-level evaluation.

3 The readout

Prompt. A decision has a state s and a question with K options $o_1, \dots, o_K$. AnyJev gives each *position* in the printed list a single-token label $\ell_1, \dots, \ell_K$: letters for a choice, Yes and No for a binary question, digits for an ordinal score. A layout π maps positions to options, so position j shows option $\pi(j)$. The prompt $x(s, \pi)$ places a fixed instruction and the state, then the question, then one line $\ell_j \cdot o_{\pi(j)}$ per position, then a line telling the model to answer with the label only. The state precedes the options, so a backend can reuse the prefix across layouts of the same question.

Decision. We run one prefill of $x(s, \pi)$ and read the next-token distribution p at the last position. Only the K label tokens can be answers, so we restrict and renormalise:

$$m(s, \pi) = \sum_{j=1}^K p(\ell_j), \quad \tilde{p}_j = \frac{p(\ell_j)}{m(s, \pi)}. \quad (1)$$

The decision is $\tilde{p}$ mapped from positions back to options. The model generates nothing, so one decision costs one prefill. Scoring single-token labels rather than the option strings also removes surface-form competition, where a correct option receives lower probability because it is spelled in more tokens or rarer ones [20].

Answer mass. The normaliser m is the share of the model’s own probability that reaches the option tokens, and it reports whether the last prompt token is the position the model would answer at. A chat template that puts something else there still yields an argmax over the label tokens, and that argmax can still be correct often enough to resemble a working readout. We report m before any accuracy and exclude a row whose m is low rather than average it in. One template fault was found this way: the harmony format used by gpt-oss ends the prompt at the assistant header, before the channel marker, which gave $m = 0.000$ while the answers still looked plausible.

Every benchmark row here has median $m \geq 0.995$ (Table 2). On the 20-option sweep three rows fall below 0.99, all on the binary injection task: 0.408 on SmoLLM2-1.7B, 0.710 on OLMo-2-7B and 0.976 on Granite-3.3-8B. We exclude those three from every statistic for that task. Appendix A prints m for every row.

Two defects. The model assigns higher probability to some labels whatever the state says, higher to Yes than to No and to earlier letters than to later ones. It also assigns higher probability to some positions in the printed list, so the decision depends on the order the caller wrote the options in. Averaged over eleven models, reversing the option list changes the argmax on 32.8% of banking20 items and on 33.4% of newsgroups items.

4 Correcting the two defects

4.1 The label prior

The label prior is a distribution $\hat{q}$ over positions that the model applies regardless of the state. We estimate it from unlabelled inputs and divide it out:

$$p^{\text{dbs}} \propto \tilde{p} / \hat{q}^\alpha, \quad (2)$$

renormalised over the K options, with $\alpha \in [0, 1]$ setting how much of the prior we remove. We take $\hat{q}$ to be the running mean of $\tilde{p}$ over the inputs the system has already scored, which is batch calibration [58], held per question and per layout. The estimate uses the model’s own outputs and no gold label. A second estimator computes $\hat{q}$ from content-free probes instead [55]; AnyJev offers it and we do not use it here.

We set $\alpha = 0.75$, taken from a separate sweep over 230 (model, question) pairs scored against gold labels, so α is the second place a label enters this system, alongside the temperature of Section 4.3. Eq. (2) also assumes the label marginal of the inputs is not extreme, and it lowers accuracy where that assumption fails.

4.2 Position bias

Position bias needs no estimator, because the layout is ours to choose. We read the same decision under the K cyclic rotations $\pi_t(j) = (j + t) \bmod K$, so that every option occupies every position once, and we average in log space [56]:

$$\bar{z}_i = \frac{1}{K} \sum_{t=0}^{K-1} \log q_t(i), \quad p^{L0} = \text{softmax}(\bar{z}), \quad (3)$$

where $q_t(i)$ is the restricted probability of Eq. (1) for option i under rotation t , before the label prior.

The log-space average makes the correction exact under a stated model of the bias. Suppose the logit for option i at position j separates as $c_i + b_j$. Then $\log q_t(i) = c_i + b_{j(t,i)} - \log Z_t$, where Z_t normalises rotation t . Summing over all K rotations sends option i through every position once, so $\sum_t b_{j(t,i)} = \sum_j b_j$ for every i , and the two remaining terms do not depend on i . Hence $\text{softmax}(\bar{z}) = \text{softmax}(c)$ exactly. Averaging probabilities instead of their logarithms does not have this property.

Two qualifications follow. The label prior is held per layout, so applying it before the average contributes a further term that cancels only when $\hat{q}$ depends on position alone; AnyJev therefore averages the pre-prior quantity. And the additive model is an approximation. Under it reversing the option list would change nothing,

and reversal still changes the answer on 0.138 of `banking20` items and 0.183 of `newsgroups` items after the average (Table 1).

Canonical order. Reading every rotation gives each option each position, but it does not fix which options are adjacent, and the options attend to one another inside the prompt. Sorting the option list by its text before rotating it removes that remaining dependence: the prompt set becomes a function of the option *set*, so any listing of the same options produces the same prompts and the same probabilities, at any rotation budget, which follows from the construction. The runs reported here rotate the caller’s listing, so we do not measure it.

4.3 Confidence

Neither correction makes the model’s confidence match its accuracy, and we did not measure a label-free procedure that detects this. One temperature, fitted on 100 to 500 labelled examples of the same question, is the correction AnyJev applies for it [19]. A single positive temperature rescales $\bar{z}$ and leaves the argmax unchanged. The level we report as *+temperature* also re-estimates $\hat{q}$ from the calibration set alone, so its accuracy differs from the debiased level by up to 0.013; we report it for calibration and read accuracy off the debiased level.

5 Reading fewer rotations

Eq. (3) requires K prefills per decision. On the 18-option routing question used for timing, that is 18 prefills where the naive readout uses one. Two changes can reduce it: make each rotation cheaper, or read fewer of them.

Section 8 reports what the first achieved. The K prompts share every token up to the option list, so an engine that caches the shared prefix already recovers most of the cost, and packing all K rotations into one forward then runs 1.20 times as fast as that path on a short state. This section reads fewer rotations instead.

Reading order. Consecutive rotations are nearly the same layout, because π_{t+1} moves every option one position from π_t . AnyJev therefore reads rotations in the order 0, $K/2$, $K/4$, $3K/4$, $K/8$, $\dots$, the van der Corput sequence scaled to K , so that a short prefix places each option in well separated positions.

When to stop. After reading a subset of rotations we hold a running average $\bar{z}$. The obvious statistic is the gap between its top two probabilities, and that gap saturates at 1. Once the average is peaked it no longer separates a settled decision from one settled by a wide margin, which is the regime the rule acts in. We use the same gap in log space,

$$\mu(\bar{z}) = \bar{z}_{(1)} - \bar{z}_{(2)}, \quad (4)$$

which is unbounded and unchanged by renormalisation of $\bar{z}$. The rule stops at the first rotation where $\mu \geq \tau$, after a minimum of two.

Choosing the threshold without labels. The guarantee we can state without labels is not about accuracy. We state it against our own full-rotation decision instead: *the budget returns the decision that reading every rotation would have returned, on at least $1 - \varepsilon$ of inputs*. That target needs only the model’s own output on unlabelled states.

Given n unlabelled states we read every rotation of each once. For a candidate τ let $d(\tau)$ count the states whose stopped decision differs from the full-rotation decision, and take the smallest τ on a grid for which

$$U(d(\tau), n, \delta) \leq \varepsilon, \quad (5)$$

where U is the Clopper–Pearson upper confidence bound [10] at confidence $1 - \delta$, with $\delta = 0.05$. If no τ satisfies Eq. (5), the question reads every rotation.

Eq. (5) is evaluated at every candidate on the grid, so a bound taken from the states that chose τ applies to each candidate separately and not to the selected one. We therefore select and verify on disjoint states: τ comes from one third of an unlabelled batch, and the bound the target must clear is computed on the other two thirds, for that single threshold. The arithmetic sets the split. At $\delta = 0.05$ a verification split of n states with one disagreement gives a bound near $4.7/n$, so a 1% target needs roughly 500 states, and 300 cannot certify 1% unless no disagreement occurs.

On four cells the procedure certifies two. Qwen2.5-7B reaches a verified bound of 0.0079 on both tasks, at 10.61 of 18 rotations and 9.01 of 20, which is 1.70 and 2.22 times fewer (Table 10). Qwen3-8B on newsgroups misses at 0.0105, and on massive_route no threshold on the grid clears the target on the selection split. Selecting and verifying on the same states, as the runs in Section 7 did, returns a smaller threshold and a larger saving, and its bound does not cover the selection.

Table 11 reports which statistic satisfies Eq. (5) at $\varepsilon = 0.01$. The log-odds margin certifies all four cells; the probability gap certifies two, and on the other two no threshold clears the target at any budget.

The cost of a round. The budget requests rotations from the engine in rounds, and every request in a batch must finish a round before the next starts. Reading one rotation per round issues the fewest forwards and the most rounds. Section 7 measures both, and the two engines rank them differently.

6 Experimental setup

Models. For the 20-option, JevBench and serving experiments, we ran thirteen instruction-tuned models from eight organisations at their released weights, in bfloat16: Qwen3 at 1.7B, 4B, 8B, 32B and 30B-A3B [49], Qwen2.5-7B [38], Llama-3.2-3B [18], Mistral-7B-v0.3 [23], Granite-3.3-8B [17], Phi-4-mini [1], SmolLM2-1.7B [5], OLMo-2-7B [44] and gpt-oss-20b [37] at its low reasoning setting. The 20-option sweep covers eleven of them from six organisations, the benchmark run eight from five. We fine-tuned, quantised and modified nothing.

Tasks. The 20-option sweep uses three tasks: **banking20**, twenty banking intents from BANKING77 [7]; **newsgroups**, the twenty classes of 20 Newsgroups [26]; and **injection**, a binary question over prompt injections. Each cell uses 300 held-out items. Timing uses an 18-option utterance routing question built from MASSIVE [14].

Benchmark. We also ran the public subset of JevBench [42], a public set of typed decisions for this class of system. It publishes 231 items in three files. Its overall score averages three difficulty tiers that partition a larger set than the public files, so we report the public subset and do not place it against leaderboard entries. We score 213 of the 231. The other 18 state an expected value that is not one of the listed options: twelve are ordinal and six are otherwise irregular, and we skip them rather than guess a mapping.

What is fitted on what. The rotation average uses no labels. Two quantities are fitted against gold labels and we name both: the prior strength α , selected once on a separate sweep (Section 4), and the temperature, fitted out of fold over items, five folds taken over the pooled items rather than within a difficulty tier. A temperature fitted on one tier alone would misfit the others. The stopping threshold of Section 5 is selected on unlabelled states of the same question.

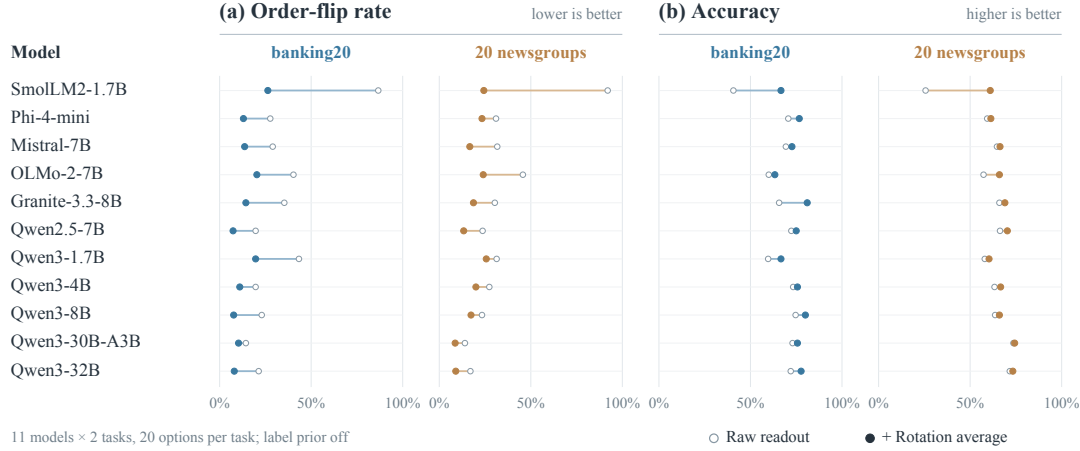


Figure 2. Rotation averaging on two 20-option tasks. Each row is one model, with separate subpanels for **banking20** and 20 newsgroups under each metric. Hollow points show the raw readout; filled points show the rotation average of Eq. (3). Left: the order-flip rate, the share of items whose answer changes when the option list is reversed. Right: accuracy. Axes are shown as percentages. All 22 model–task pairs have a lower order-flip rate and higher accuracy after averaging. The label prior is switched off, so the changes are due to rotations alone.

Settings and hardware. The main tables read every rotation, combine in log space, and rotate the caller’s listing rather than a canonical one. The 20-option sweep applies the label prior of Eq. (2); the benchmark run does not, so its debiased column isolates the rotations. These experiments used one H100 NVL, PyTorch 2.5.1 with CUDA 12.4 and Transformers 4.55.4 [46]. The serving comparison additionally used vLLM 0.7.0 [25] on the same host, measured serially on an otherwise idle machine, because an earlier pair of runs that shared the host returned a different ratio.

Additional full-split evaluation. We separately evaluate the default BEV Decision Mix test split on two AnyJev base models and four published baseline interfaces. Its readouts, calibration protocol and software settings are specified in Appendix E; the temperature fit and hardware configuration above do not describe that separate evaluation.

7 Results

7.1 Order dependence falls on every model tested

The rotations lower the order-flip rate on 11 of 11 models on both 20-option tasks, from 0.328 to 0.138 on **banking20** and from 0.334 to 0.183 on **newsgroups** (Table 1, Figure 2). They raise accuracy on 11 of 11 models on both, by 7.2 and 5.9 points on the means. An exact sign test over the eleven models puts each of those counts at $p = 4.9 \times 10^{-4}$.

The size of the effect depends on the number of options. On the binary injection task the rotations remove order dependence completely, from 0.180 to 0.000 on the eight rows that pass the answer-mass gate, and they do not raise accuracy: they win on 3 of the 7 rows that are not ties, at $p = 0.77$. With two options there are two positions, and little position bias to average away.

Calibration moves under the temperature rather than under the rotations. The rotations and the prior take the calibration error from 0.251 to 0.223 on **banking20**, and the temperature takes it to 0.088.

	banking20	20 newsgroups	injection
	$K = 20$	$K = 20$	$K = 2$
<i>Accuracy</i>			
raw readout	0.665	0.609	0.720
+ rotations	0.737	0.668	0.733
+ rotations + label prior	0.752	0.675	0.762
<i>Order-flip rate</i>			
raw readout	0.328	0.334	0.180
+ rotations	0.138	0.183	0.000
<i>Calibration error</i>			
raw readout	0.251	0.276	0.202
+ rotations + label prior	0.223	0.285	0.170
+ temperature	0.088	0.099	0.095
Models passing the answer-mass gate	11 of 11	11 of 11	8 of 11
Rotations raise accuracy on	11/11	11/11	3/7
Rotations lower the flip rate on	11/11	11/11	8/8

Table 1. Means over the models that pass the answer-mass gate, 300 items per cell. The lower block counts models rather than items. Under a coin flip, 11 of 11 occurs with probability 4.9×10^{-4} (exact sign test, one-sided). The injection accuracy row is 3 of 7 after dropping one tie, at $p = 0.77$.

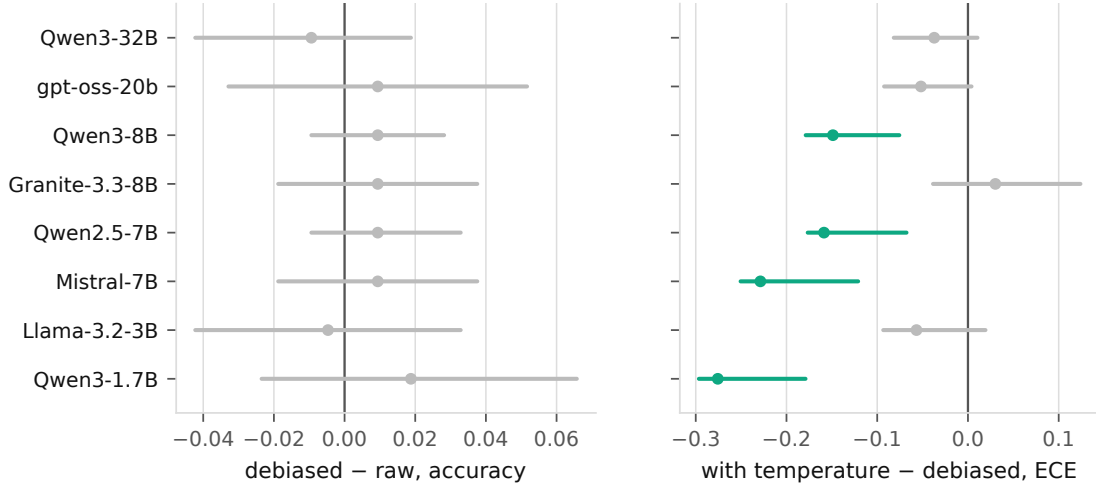


Figure 3. Differences on the JevBench public subset with 95% bootstrap intervals over 2 000 resamples, paired over items, one row per model. Left: debaised minus raw accuracy; every interval contains zero. Right: the change in calibration error from adding the temperature; four of the eight intervals exclude zero, and all four are improvements. Coloured intervals exclude zero.

7.2 The benchmark

Accuracy on this subset spans 0.549 for Qwen3-1.7B to 0.798 for Qwen3-32B (Table 2). The spread belongs to the models. The debiasing does not change accuracy here by a measurable amount: no per-model interval excludes zero, and pooling all $8 \times 213 = 1704$ decisions gives a difference of $+0.006$ with a 95% interval of $[-0.004, +0.018]$.

Section 4 predicts that the effect grows with the number of options, and this benchmark has few. Its items have 2 to 6 options, 3.6 on average over the 213 scored. Table 8 splits the pooled decisions by option

Model	Answer mass	Raw	Debiased	ECE raw	ECE debiased	ECE + temp.
Qwen3-32B	1.000	0.798	0.789	0.140	0.128	0.091
Qwen3-8B	1.000	0.704	0.714	0.278	0.263	0.114
gpt-oss-20b	0.999	0.704	0.714	0.140	0.154	0.102
Granite-3.3-8B	0.999	0.690	0.700	0.280	0.270	0.301
Qwen2.5-7B	0.996	0.676	0.685	0.259	0.243	0.085
Mistral-7B	0.999	0.615	0.624	0.322	0.333	0.104
Llama-3.2-3B	0.999	0.563	0.559	0.174	0.163	0.107
Qwen3-1.7B	1.000	0.549	0.568	0.395	0.380	0.105

Table 2. JevBench public subset, 213 items scored per model. Answer mass is the normaliser of Eq. (1). ECE is expected calibration error over 15 equal-mass bins. Intervals for the differences are in Figure 3 and Table 7.

count. The $K = 6$ interval excludes zero, but it is one of five intervals. Were they independent, the chance that at least one of five 95% intervals excludes zero under the null would be 0.23; they share models and the resampling is clustered, so that figure is indicative rather than exact. Testing the trend once, by regressing the per-decision difference on K and bootstrapping the slope clustered by item, gives +0.0059 accuracy per additional option with an interval of $[-0.0012, +0.0132]$. The sign matches the prediction and the sample does not resolve it. We therefore report no accuracy effect on this benchmark.

The temperature lowers calibration error on seven of the eight models and raises it on one. Four of the eight intervals exclude zero, all of them improvements: Qwen3-8B, Qwen2.5-7B, Mistral-7B and Qwen3-1.7B, between -0.149 and -0.276 . On Granite-3.3-8B the error rises from 0.270 to 0.301, with an interval that contains zero. One global temperature cannot correct a model whose confidence is wrong in different directions on different items.

7.3 Full BEV Decision Mix evaluation

On the full test split of the default BEV Decision Mix configuration [3], all systems are scored on the same 46 320 decisions (Appendix E, Table 14). With the label prior off, rotation averaging reaches 68.01% on Qwen3.5-9B and 67.59% on Qwen3-32B, compared with 67.85% and 67.03% for their raw readouts. Bespoke-Nimble-9B reaches 70.57%; its 2.56-point lead over the 9B rotation average is an observed system difference, not an isolated estimate of the effect of fine-tuning. Adding the batch prior at $\alpha = 0.75$ lowers accuracy to 65.01% and 65.55%, respectively, although it improves the 9B model’s ECE from 0.0490 to 0.0418. Thus the prior’s accuracy benefit does not transfer to this evaluation. The evaluated affine-map L2-mono variant reads block 29 of 32 and scores 67.24%, 0.61 points below the same model’s full-depth, single-forward readout; latency was not measured.

7.4 Serving cost

On the 18-option routing question, Eq. (5) returns a threshold of 6.0 from 600 unlabelled states, with one disagreement and a Clopper–Pearson upper bound of 0.0079 against a 1% target. Reading two rotations per engine call, the budget then reads 7.3 of the 18 and returned the full-rotation decision on 98.7% of held-out decisions (Table 3). That is a disagreement rate of 0.013, or 4 decisions of 300. This run selected its threshold and took its bound from the same states, so the bound does not cover the selection, and the rate is the one to read. Section 5 reports the same cell under selection and verification on disjoint states: the threshold rises from 6.0 to 10.25, the rotations read from 7.3 to 10.6, and the bound becomes one a deployment can rely on. We quote 6.0 here because it is the threshold the serving run used.

The round size matters as much as the rotation count, and the two engines rank it differently. On vLLM, reading one rotation per round issues the fewest rotations, 7.2 of 18, and reaches 22.1 decisions per second.

Engine	Readout	Decisions/s	Speed-up	Rotations read	Accuracy	Agreement
vLLM	all 18 rotations	16.7	1.00	18.0	0.697	1.000
	budget, wave 1	22.1	1.32	7.2	0.703	0.987
	budget, wave 2	37.2	2.22	7.3	0.703	0.987
	budget, wave 4	31.8	1.90	9.3	0.703	0.987
	budget, wave 6	29.2	1.74	10.9	0.700	0.987
Transformers	all 18 rotations	7.0	1.00	18.0	0.697	1.000
	budget, wave 1	16.5	2.34	7.2	0.703	0.987
	budget, wave 2	16.5	2.34	7.4	0.703	0.987
	budget, wave 4	13.1	1.87	9.2	0.703	0.987
	budget, wave 6	11.4	1.62	10.8	0.700	0.990

Table 3. 300 decisions of an 18-option question on one H100. The wave is how many rotations a decision requests per engine call. Accuracy spans 0.697 to 0.703 across the rows, and agreement is measured against the full-rotation decision. Speed-up is against the all-rotations row of the same engine.

Reading two per round issues 7.3 and reaches 37.2, which is 2.22 times the all-rotations rate. At four per round the rate falls to 31.8. On Transformers the first two rounds sizes are level at 16.5 decisions per second, 2.34 times the all-rotations rate, and larger rounds are slower.

We also measured whether the decision can be read from a truncated forward, by fitting a label-free affine map from an intermediate block into the final basis [4]. At a 0.98 agreement target six of the eight models save no blocks. Appendix D reports the depths.

8 What did not work

Each mechanism below does what its design says, to the precision quoted, and a measurement rather than a fault stopped each one.

A question-agnostic head. A closed-form head fitted on one question’s gold labels against the model’s hidden states scores 0.13 above the debiased readout at the same depth, and it is fitted per question. Leaving out one of twenty questions at a time and fitting on the other nineteen, four extraction variants, five depths and four functional forms all score below the raw readout on the held-out question. The best reaches 0.580, against 0.635 for the debiased readout and 0.628 for the raw one, and its flip rate under reversal is 0.17 to 0.30, against 0.07 for the raw readout on the same questions. The signal is present per question; a shared linear rule that reaches an unseen question is not.

Reusing the question block’s cache. The question and its options are identical across requests to a given endpoint, but they follow the state, so no prefix cache reaches them. On a short state they are most of the prompt: 159 fixed tokens of which 42 precede the state, against a median routing utterance of 7 tokens. Splicing the cached block into a new request and rotating its positions is exact to 7.0×10^{-5} in fp32 across all 28 layers. The reused block then reproduces the previous state’s decision in the new request.

Making the options a set. The K rotations exist because the options are written in sequence, so option 5 attends to options 1 through 4 and option 1 to none of them. A mask in which every option span sees the prefix and itself, each starting at the same position index, makes the options exchangeable by construction: shuffling the spans moves the output by 3.7×10^{-6} under it, against 2.2×10^{-3} under the causal mask. Accuracy then falls to chance. On Qwen2.5-7B and the 18-option routing question it scores 0.042 against 0.633 for the

rotation average, where chance is 0.056. Masking the options from each other while leaving their positions alone already scores 0.092, so the loss comes from the blinding rather than the shared positions. The model uses the other options when it selects among them.

Packing the rotations into one forward. Laying the K option blocks end to end, each at the same starting position index and masked to see only the prefix and itself, reproduces K independent prefills to 2.2×10^{-5} and returns an identical decision. It runs 5.9 times as fast as K independent prefills, and 1.20 times as fast as the shared-prefix path an engine already provides on a 10-token state, 1.11 times on a 200-token state. This is why Section 5 reduces the number of rotations instead of their unit cost.

9 Limitations

- **One prefill sets a ceiling.** The readout generates nothing, so it cannot reason. On the hardest tier of the JevBench subset the eight models span 0.343 to 0.590.
- **JevBench has few options.** Its scored items average 3.6 options, the range where the rotation average has least to remove. We do not have a public many-option benchmark, and one would test the mechanism where it operates.
- **Eighteen JevBench items are unscored.** Twelve of the 231 public items are ordinal, where the expected value names a level rather than one of the listed options.
- **The 20-option results have no intervals.** Those runs record aggregates rather than per-item outcomes, so the strongest statement available from them is the sign test over eleven models.
- **The intervals are over items, not over runs.** They state what a different sample of items would have given. They do not cover seed, host, or the effect of batch composition on `bfloat16` logits.
- **The threshold is selected and certified on the same sample.** The bound therefore covers each candidate on the grid and not the selected one. On the serving run the held-out rate was 0.013 on 300 decisions, whose 95% interval $[0.004, 0.034]$ contains the 0.01 target, so that run neither confirms nor refutes it. Selecting on one unlabelled split and measuring on a second would settle it; we did not do that here.
- **The efficiency and accuracy measurements use different tasks.** The JevBench subset’s K is 2 to 6, so a rotation budget has little to save there.
- **Hidden-state readouts need a local backend.** The affine map of Appendix D and the per-question head need hidden states, which a hosted API does not expose. The readout of Section 4 needs only next-token log-probabilities.

10 Conclusion

The decision capability measured here is the base model’s. What AnyJev supplies is a readout with the label prior divided out and the position term averaged away, at one prefill per rotation and no parameter changes. On both 20-option tasks that readout lowered the order-flip rate on every model we tested, from 0.33 to 0.138 and 0.183, and a stopping rule selected against its own full-rotation decision reduced the number of rotations read. AnyJev is open source.

Status. This is an early report on work that is still in development. It covers the readout, the corrections and their serving cost; the labelled head is not evaluated. The evidence includes the 213-item JevBench public subset, three tasks of our own, and 46 320 decisions from the default BEV Decision Mix test split, including comparisons with published decision systems. On that full split, the best AnyJev accuracy uses rotations

without the label prior; the prior’s benefit is therefore task-dependent. These are early measurements, not a settled ranking of decision systems.

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A Per-model results on the 20-option tasks

Table 1 reports means. The three tables below give each model separately. *mass* is the answer mass of Eq. (1), in bold where it falls below the 0.99 gate. *rot.* is the rotation average of Eq. (3) with the label prior switched off, *prior* adds Eq. (2), and *temp.* adds the fitted temperature. *cov@5%* is the share of items the system can answer, taken in confidence order, before the error rate on the answered set passes 5%.

Model	mass	raw	+rot.	+prior	ECE raw	ECE +temp.	flip raw	flip +rot.	cov@5% raw	cov@5% +temp.
Granite-3.3-8B	1.000	0.657	0.810	0.807	0.293	0.052	0.353	0.143	0.033	0.523
Qwen3-8B	1.000	0.747	0.800	0.803	0.240	0.095	0.230	0.077	0.077	0.520
Phi-4-mini	0.999	0.707	0.767	0.783	0.185	0.065	0.277	0.130	0.017	0.480
Qwen3-32B	1.000	0.720	0.777	0.780	0.227	0.064	0.213	0.080	0.017	0.443
Qwen3-30B-A3B	1.000	0.730	0.757	0.770	0.249	0.086	0.143	0.103	0.277	0.470
Qwen3-4B	1.000	0.733	0.757	0.760	0.254	0.098	0.197	0.110	0.103	0.423
Qwen2.5-7B	0.998	0.723	0.750	0.757	0.237	0.070	0.197	0.073	0.180	0.223
Mistral-7B	1.000	0.693	0.727	0.740	0.287	0.113	0.290	0.137	0.100	0.347
SmolLM2-1.7B	0.992	0.407	0.667	0.717	0.219	0.102	0.867	0.263	0.093	0.360
OLMo-2-7B	0.999	0.600	0.633	0.680	0.185	0.105	0.403	0.203	0.003	0.037
Qwen3-1.7B	1.000	0.597	0.667	0.670	0.383	0.121	0.433	0.197	0.027	0.260

Table 4. banking20, $K = 20$, 300 items per model. All eleven rows pass the gate.

Model	mass	raw	+rot.	+prior	ECE raw	ECE +temp.	flip raw	flip +rot.	cov@5% raw	cov@5% +temp.
Qwen3-30B-A3B	1.000	0.737	0.743	0.740	0.242	0.096	0.140	0.087	0.303	0.003
Qwen3-32B	1.000	0.717	0.733	0.737	0.236	0.063	0.170	0.090	0.557	0.410
Qwen2.5-7B	0.998	0.663	0.703	0.710	0.273	0.096	0.237	0.133	0.047	0.407
Granite-3.3-8B	1.000	0.660	0.690	0.693	0.328	0.111	0.303	0.187	0.000	0.183
OLMo-2-7B	0.999	0.573	0.660	0.683	0.203	0.047	0.457	0.240	0.283	0.290
Mistral-7B	0.995	0.647	0.663	0.673	0.315	0.096	0.317	0.167	0.007	0.073
Qwen3-4B	1.000	0.633	0.667	0.667	0.341	0.138	0.273	0.200	0.000	0.157
Qwen3-8B	0.999	0.637	0.660	0.660	0.334	0.138	0.233	0.173	0.013	0.237
SmolLM2-1.7B	0.994	0.257	0.610	0.637	0.097	0.089	0.920	0.243	0.003	0.137
Phi-4-mini	0.999	0.593	0.613	0.617	0.272	0.110	0.310	0.233	0.080	0.217
Qwen3-1.7B	1.000	0.580	0.603	0.610	0.394	0.106	0.313	0.257	0.010	0.000

Table 5. newsgroups, $K = 20$, 300 items per model. All eleven rows pass the gate.

Model	mass	raw	+rot.	+prior	ECE raw	ECE +temp.	flip raw	flip +rot.	cov@5% raw	cov@5% +temp.
Qwen3-32B	0.999	0.857	0.833	0.827	0.062	0.067	0.073	0.000	0.550	0.483
Phi-4-mini	0.994	0.707	0.790	0.807	0.133	0.106	0.427	0.000	0.017	0.010
Qwen3-4B	1.000	0.683	0.783	0.797	0.298	0.081	0.203	0.000	0.117	0.100
Qwen2.5-7B	0.999	0.737	0.720	0.790	0.188	0.049	0.070	0.000	0.367	0.380
Qwen3-30B-A3B	0.995	0.730	0.730	0.757	0.248	0.090	0.103	0.000	0.410	0.410
Mistral-7B	0.995	0.697	0.707	0.733	0.167	0.069	0.333	0.000	0.017	0.143
Qwen3-8B	1.000	0.693	0.677	0.700	0.288	0.161	0.060	0.000	0.313	0.313
Qwen3-1.7B	0.992	0.657	0.627	0.683	0.231	0.135	0.170	0.000	0.000	0.000
SmolLM2-1.7B	0.408	0.593	0.593	0.670	0.215	0.095	0.000	0.000	0.030	0.000
Granite-3.3-8B	0.976	0.627	0.637	0.647	0.302	0.103	0.100	0.000	0.000	0.003
OLMo-2-7B	0.710	0.593	0.593	0.593	0.367	0.083	0.003	0.000	0.000	0.000

Table 6. injection, $K = 2$, 300 items per model. Three rows fall below the answer-mass gate and are excluded from every statistic reported for this task. The rotation average removes order dependence on every row that passes.

B Benchmark detail

Model	raw	debiased	diff.	95% CI	ECE deb.	ECE +temp.	diff.	95% CI
Qwen3-32B	0.798	0.789	-0.009	[-0.042, +0.019]	0.128	0.091	-0.037	[-0.082, +0.010]
Qwen3-8B	0.704	0.714	+0.009	[-0.009, +0.028]	0.263	0.114	-0.149	[-0.179, -0.076]
gpt-oss-20b	0.704	0.714	+0.009	[-0.033, +0.052]	0.154	0.102	-0.052	[-0.092, +0.004]
Granite-3.3-8B	0.690	0.700	+0.009	[-0.019, +0.038]	0.270	0.301	+0.030	[-0.038, +0.124]
Qwen2.5-7B	0.676	0.685	+0.009	[-0.009, +0.033]	0.243	0.085	-0.159	[-0.177, -0.068]
Mistral-7B	0.615	0.624	+0.009	[-0.019, +0.038]	0.333	0.104	-0.229	[-0.251, -0.121]
Qwen3-1.7B	0.549	0.568	+0.019	[-0.023, +0.066]	0.380	0.105	-0.276	[-0.297, -0.179]
Llama-3.2-3B	0.563	0.559	-0.005	[-0.042, +0.033]	0.163	0.107	-0.057	[-0.093, +0.019]

Table 7. Per-model differences with 95% bootstrap intervals over 2 000 resamples, paired over items. The temperature is refitted inside every resample. Reusing a fit made on the full sample would place the whole sample inside every draw and narrow the interval.

Options K	Items	Decisions	Raw	Debiased	Difference	95% CI
2	74	592	0.677	0.676	-0.002	[-0.017, +0.015]
3	15	120	0.333	0.317	-0.017	[-0.050, +0.017]
4	53	424	0.649	0.665	+0.017	[-0.012, +0.045]
5	55	440	0.805	0.811	+0.007	[-0.016, +0.030]
6	16	128	0.461	0.492	+0.031	[+0.008, +0.062]
Slope of the difference on K					+0.0059	[-0.0012, +0.0132]
Pooled over all 1704 decisions					+0.0065	[-0.0041, +0.0176]

Table 8. The difference against the number of options, all eight models pooled. Resampling is clustered by item, so a drawn item contributes all eight models’ decisions. The final two rows replace those five interval tests with a single test of the trend.

Model	easy (48)	original (72)	hard (111)
Qwen3-32B	1.000	0.967	0.590
Qwen3-8B	1.000	0.817	0.524
gpt-oss-20b	1.000	0.850	0.505
Granite-3.3-8B	1.000	0.900	0.448
Qwen2.5-7B	1.000	0.867	0.438
Mistral-7B	1.000	0.717	0.400
Llama-3.2-3B	0.917	0.650	0.343
Qwen3-1.7B	1.000	0.600	0.352

Table 9. Accuracy by difficulty tier of the public subset. These are the three public files, not the partition the benchmark’s overall score uses. The counts in the header are the published items per file; the accuracies are over the scored items only, which are 48 easy, 60 original and 105 hard, because the 18 unscored items fall in the second and third files.

C Rotations and the certificate

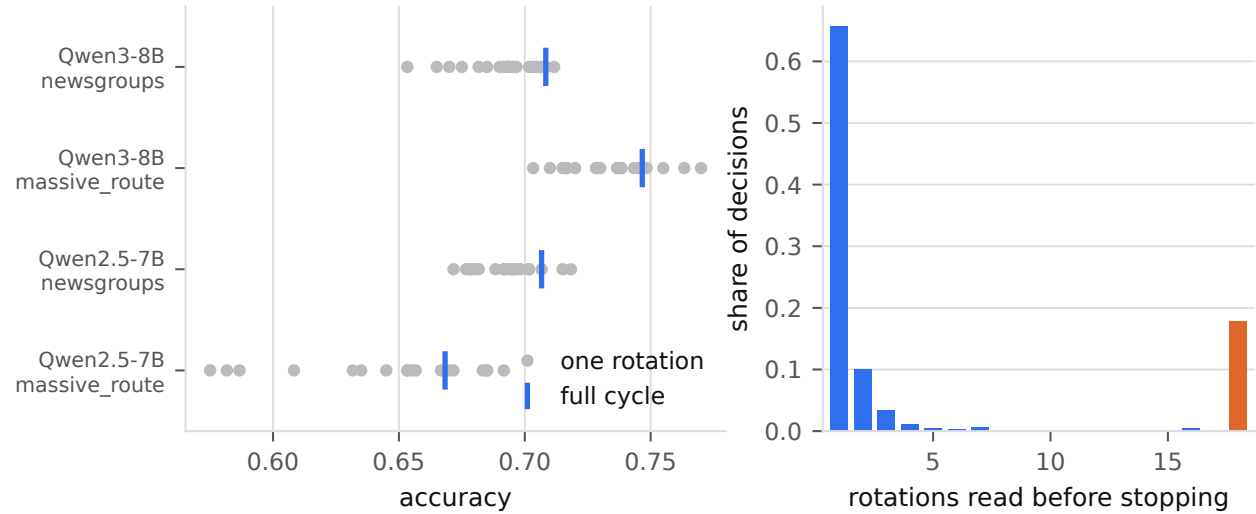


Figure 4. Left: every cyclic rotation scored on its own, against the full-rotation average, on four (model, task) cells. On all four the average falls at or below the best single rotation, by 0.3 to 2.3 points, and above the worst by 3.5 to 9.3 points. Right: where the budget stops on one 18-option cell, over 600 decisions, for a rule with no minimum-rotation floor; the shipped default reads two rotations before it may stop, which moves the leftmost bar.

The left panel states what the rotations do. They do not raise accuracy above the best single rotation. They remove the caller’s ability to select a poor one, and they reduce how far the result depends on the order the options were written in.

D Reading a truncated forward

Truncating the forward at block b leaves the residual stream outside the basis that the output head reads. We fit an affine map from block b into that basis, targeting the model’s own full-depth state on unlabelled inputs, which is a tuned lens [4]. We select the depth on a calibration corpus disjoint from every evaluation set. Reading the truncated state through the output head without the map is the logit lens [36]. The quantity throughout is agreement with the model’s own full-depth decision, not accuracy.

Across the eight models the transition is sharp. Seven never exceed 0.47 agreement at any depth below

Model	Task	K	Threshold	Verify rate	Verify bound	Clears 1%	Rotations	vs. K
Qwen2.5-7B	massive_route	18	10.25	0.0017	0.0079	yes	10.61	1.70×
Qwen2.5-7B	newsgroups	20	9.50	0.0017	0.0079	yes	9.01	2.22×
Qwen3-8B	massive_route	18	—	—	—	—	—	—
Qwen3-8B	newsgroups	20	9.00	0.0033	0.0105	no	5.31	3.77×

Table 10. The certificate with the threshold selected on one third of an unlabelled batch and the bound computed on the other two thirds, for that one threshold. The verify bound carries no selection multiplicity. Two of the four cells clear a 1% target this way; on massive_route with Qwen3-8B no threshold on the grid cleared it on the selection split.

Stopping statistic	Cells certified	Mean rotations saved
log-odds margin alone	4 of 4	3.46×
log-odds margin + unanimity	4 of 4	2.91×
probability gap alone	2 of 4	3.19×
probability gap + unanimity	2 of 4	2.64×

Table 11. Which stopping statistic satisfies Eq. (5) at $\varepsilon = 0.01$, over four (model, task) cells. A rule with fewer certified cells did not run worse: no threshold on the grid cleared the target on those cells at any budget.

the crossing, and then reach 0.90 in a single measured step. The depth at which that happens ranges from 66% to 100% of blocks and does not follow parameter count. At a 0.98 target six of the eight models save no blocks. The two that do are Mistral-7B at 66% of blocks and Qwen3-8B at 89%.

Part of what the table charges to truncation is the map’s own error. At full depth no truncation occurs, so any agreement below 1.000 in the last column is the map alone. It is 0.000 on Mistral-7B and 0.007 on Qwen3-8B, the two models that save blocks, and 0.040 on Llama-3.2-3B, which is why that model reaches no depth at the 0.98 target.

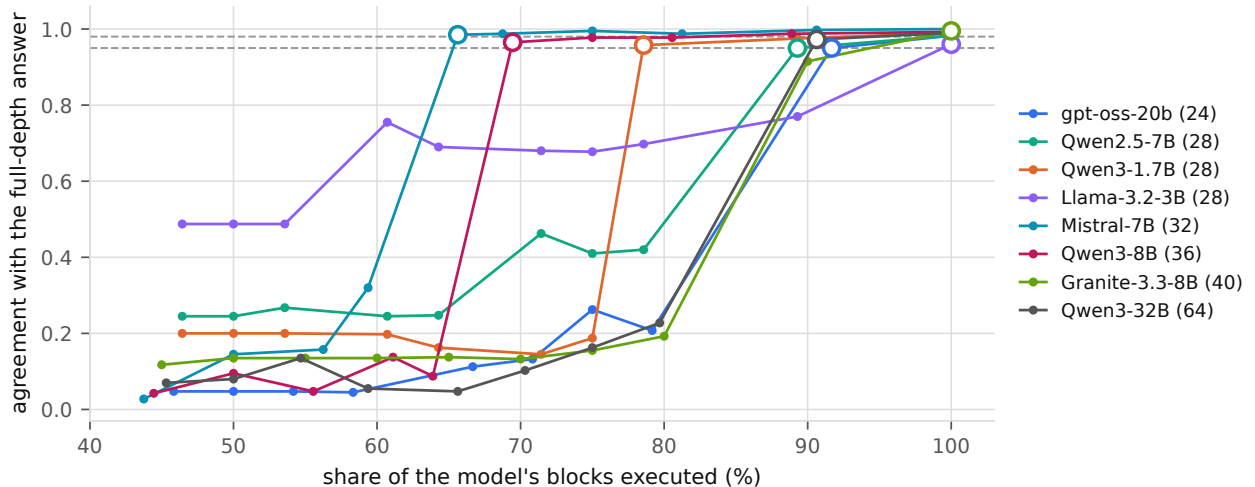


Figure 5. Agreement against depth, eight models. Markers are the depths we ran, and the curve is drawn through them rather than smoothed, because the intermediate depths were not measured. Hollow markers show the first depth reaching 0.95. Dashed lines mark 0.95 and 0.98.

Stopping statistic	Model	Task	K	Threshold	Rotations	vs. K	Disagreement
log-odds margin alone	Qwen2.5-7B	massive_route	18	4.5	6.12	2.94×	0.0083
	Qwen2.5-7B	newsgroups	20	6.0	6.00	3.34×	0.0000
	Qwen3-8B	massive_route	18	9.5	5.37	3.36×	0.0000
	Qwen3-8B	newsgroups	20	8.25	4.75	4.21×	0.0033
log-odds margin + unanimity	Qwen2.5-7B	massive_route	18	4.25	7.96	2.26×	0.0083
	Qwen2.5-7B	newsgroups	20	6.0	6.63	3.02×	0.0000
	Qwen3-8B	massive_route	18	9.5	6.38	2.82×	0.0000
	Qwen3-8B	newsgroups	20	8.25	5.63	3.55×	0.0017
probability gap alone	Qwen2.5-7B	massive_route	18	0.97	5.87	3.07×	0.0083
	Qwen2.5-7B	newsgroups	20	0.995	6.05	3.30×	0.0000
probability gap + unanimity	Qwen2.5-7B	massive_route	18	0.965	7.89	2.28×	0.0083
	Qwen2.5-7B	newsgroups	20	0.995	6.69	2.99×	0.0000

Table 12. Every cell each stopping rule certifies at $\varepsilon = 0.01$, with its threshold and the rotations it then reads. Rules with two rows did not certify the other two cells at any threshold on the grid. These thresholds come from a different calibration sample than the serving run of Section 7, which is why the same cell appears there at 6.0.

Model	Blocks	Depth at 0.98	Depth at 0.95	Before crossing	At crossing	Agreement at full depth
Mistral-7B	32	66%	66%	0.320 at 59%	0.985 at 66%	1.000
Qwen3-8B	36	89%	69%	0.087 at 64%	0.965 at 69%	0.993
Qwen3-1.7B	28	100%	79%	0.188 at 75%	0.958 at 79%	0.990
Qwen2.5-7B	28	100%	89%	0.420 at 79%	0.950 at 89%	0.983
Qwen3-32B	64	100%	91%	0.228 at 80%	0.973 at 91%	0.990
gpt-oss-20b	24	100%	92%	0.207 at 79%	0.950 at 92%	0.983
Granite-3.3-8B	40	100%	100%	0.193 at 80%	0.915 at 90%	0.995
Llama-3.2-3B	28	not reached	100%	0.770 at 89%	0.960 at 100%	0.960

Table 13. Agreement with the model’s own full-depth decision, for a readout taken at a fraction of the blocks and mapped into the final basis. *Before crossing* and *at crossing* are the last depth measured below 0.90 agreement and the first at or above it.

E Full BEV Decision Mix results

We evaluate the original default configuration of BEV Decision Mix [3]. Its 24 386 test rows expand to 46 320 state-question decisions: 18 808 choices among 2–10 options, 17 572 yes/no decisions (noul), and 9 940 ordinal scores with 3–7 levels. Table 14 summarizes all readouts evaluated on the full test split. The sampled Qwen2.5-7B depth experiment is excluded from this comparison.

Table 14. All full-split BEV Decision Mix results. Accuracy and its intervals are in percent; ECE is on a 0–1 scale. Higher accuracy and lower ECE are better. Bold marks the best measured system in each metric, including ties.

System / readout	Overall ↑	95% CI	Choice <i>n</i> = 18,808	Yes/no <i>n</i> = 17,572	Ordinal <i>n</i> = 9,940	ECE ↓ (shard mean)
<i>Reference rules (not learned systems)</i>						
Uniform random (expected)	34.29	—	25.77	50.00	22.63	—
Majority by question (test oracle)	54.28	—	41.42	73.57	44.51	—
<i>Published baseline interfaces</i>						
Bespoke-Nimble-9B	70.57	[70.16, 70.99]	73.60	81.06	46.33	0.0889
Laya typed-decisions	56.20	[55.75, 56.65]	55.94	72.97	27.05	0.0444
Qwen3-Reranker-0.6B	40.50	[40.05, 40.95]	52.47	37.46	23.22	0.2249
NanoJev	32.99	[32.56, 33.42]	35.43	34.57	25.55	0.2151
<i>AnyJev on Qwen3.5-9B</i>						
Raw (one forward)	67.85	[67.43, 68.28]	67.46	81.20	45.00	0.0699
L0: rotations, prior off	68.01	[67.59, 68.43]	67.69	81.37	45.00	0.0490
L0: batch prior, $\alpha = 0.75$	65.01	[64.58, 65.45]	66.37	74.60	45.50	0.0418
Rotations + batch prior, $\alpha = 1.0$	62.21	[61.77, 62.65]	63.70	70.69	44.42	0.0565
Rotations + content-free, $\alpha = 0.5$	64.45	[64.01, 64.88]	69.39	71.86	41.99	0.0839
Rotations + content-free, $\alpha = 0.75$	62.10	[61.66, 62.54]	69.52	66.38	40.48	0.1122
Rotations + content-free, $\alpha = 1.0$	58.95	[58.50, 59.40]	68.55	60.68	37.75	0.1483
L2-mono (affine map, block 29/32)	67.24	[66.81, 67.66]	66.80	81.31	43.20	0.0888
<i>AnyJev on Qwen3-32B</i>						
Raw (one forward)	67.03	[66.60, 67.46]	69.49	75.45	47.51	0.1982
L0: rotations, prior off	67.59	[67.17, 68.02]	70.40	75.96	47.51	0.1907
L0: batch prior, $\alpha = 0.75$	65.55	[65.12, 65.99]	68.16	72.99	47.48	0.1899
Rotations + batch prior, $\alpha = 1.0$	62.80	[62.36, 63.24]	64.95	70.10	45.86	0.2063
Rotations + content-free, $\alpha = 0.5$	63.51	[63.07, 63.95]	70.39	66.56	45.10	0.2388
Rotations + content-free, $\alpha = 0.75$	61.62	[61.18, 62.07]	70.06	62.39	44.31	0.2613
Rotations + content-free, $\alpha = 1.0$	59.24	[58.79, 59.68]	68.88	58.20	42.82	0.2854

Notes. Every measured row contains 46 320 decisions. Accuracy pools correct counts across shards. ECE uses 15 equal-mass bins within each shard; the reported value is their mean (four shards for AnyJev and Nimble, one for the other baselines). It is not pooled ECE. Wilson intervals treat decisions as independent and do not account for repeated states or questions; they are descriptive, not a test of between-system differences. The majority reference chooses its class from test labels within each question-ID/option-count group and is an oracle diagnostic, not a deployable baseline. Uniform random is its analytic expected accuracy. Neither reference has a reported confidence interval or ECE.

Readout protocol. Raw reads the listed order once. L0 with the prior off averages every cyclic rotation in log space; ordered score levels retain their original order, so their raw and rotation results coincide. The batch prior subtracts the question-wise mean option log-score within each evaluation shard at the displayed strength, for groups of at least 20 items; it uses test inputs but no test labels. The content-free alternative subtracts scores from three neutral inputs (empty text, N/A, and [MASK]). Both are applied after rotation averaging. No temperature is fitted, and no row represents L1 or the supervised L2 head.

Systems and interpretation. AnyJev keeps the released base-model weights frozen. Nimble [6], Laya [27] and NanoJev [35] use their published prediction interfaces; Qwen3-Reranker-0.6B scores each option separately using its yes/no reranking template. Question labels are excluded from model inputs. The dataset stores each label inside the same object as its question text, so the one run made before that exclusion was in place, Laya, was repeated with it; the two runs agree on all 46 320 decisions. None of the four baselines was trained on this dataset: Nimble on its own synthetic set, Laya on LocalLLaMA/typed-decisions, NanoJev on recorded expert episodes from four game environments, and Qwen3-Reranker on retrieval data. Their results here measure transfer rather than performance on the workloads they were built for. Nimble uses a Qwen3.5-9B LoRA adapter, so it shares AnyJev’s base architecture but not its final weights or prompt. Its 70.57% overall accuracy exceeds the best AnyJev result of 68.01%; the latter has lower ECE (0.0490 versus 0.0889). The two systems are close on yes/no decisions (81.37% for AnyJev versus 81.06% for Nimble), while Nimble leads on choices (73.60% versus 67.69%). All measured systems have their lowest accuracy on ordinal scores. Every tested prior correction lowers overall accuracy relative to rotation averaging alone on its respective base model, even where ECE improves.

The evaluated L2-mono variant. This run fits an affine map to the model’s own full-depth states on 1 500 training-split prompts, without gold labels. The first two thirds fit the map; the remaining third selects ridge strength and the earliest tested block reaching 95% agreement with the full-depth, single-forward decision. It selects block 29/32, with 95.6% calibration agreement. Its 67.24% test accuracy is 0.61 points below the raw single forward and 0.77 points below the rotation average. The 9.4% reduction in block count is not a measured latency improvement, and the calibration agreement is not a test-set guarantee.

Scope. These measurements use two AnyJev bases on one dataset configuration. We use Transformers 4.55 for Qwen3-32B and Qwen3-Reranker-0.6B, and 5.17 for Qwen3.5-9B, Nimble and NanoJev; Laya uses its own package. AnyJev prompts are capped at 2 048 tokens, whereas the Nimble interface supports up to 8 192, so this is a comparison of evaluated systems rather than an equal-context ablation. No cross-system latency or memory comparison was measured.